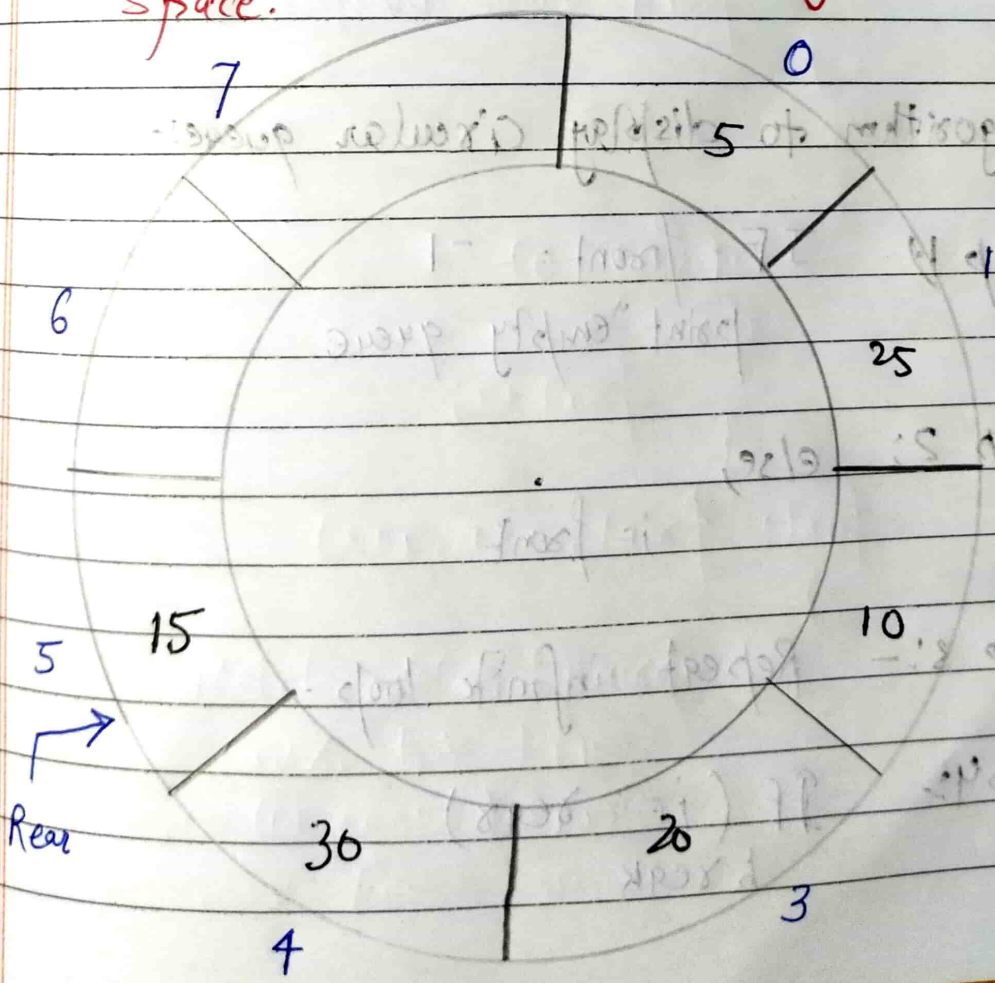
Circular Queue

A circular Queue is a Linear data structure that follows the FIFO (First In, First Out) principle, but unlike a regular queue, it connects the last position back to the first, forming a circle.

→ Key Concept of Circular Queue:-

- Front : points to the first element.
- Rear : points to the last element.
- When rear reaches the end, it wraps around to the begining if there's space.



Algorithm:- To create circular Queue:-

Step 1:- IF n > Max
print "queue overflow"

Step 2:- else, if n! = 0
front ++;

Step 3:- Under for loop condition

Step 4:- rear = (rear+1) % max

Step 5:- queue [rear] = val

Step 6:- Exit

Algorithm to display circular queue:-

Step 1:- IF front == -1
print "empty queue

Step 2:- else,
i = front

Step 3:- Repeat infinite loop

Step 4:- IF (i == rear)
break

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Step 5:- i = (i + 1) % max

Step 6 :- Exit;

create &

→ // c program to display Circular Queue;

```
#include<stdio.h>
#define max 5

int queue[max], front=-1, rear=-1;

void Create_Circular_queue()
{
    int i,val,n;
    printf("enter size of an queue");
    scanf("%d",&n);

    if(n>max)
    {
        printf("queue overflow");
    }
    else
    {
        if(n!=0)
        {
            front++;
        }
    for(i=front; i<n; i++)
    {
        printf("enter data\n");
        scanf("%d",&val);
        rear=(rear+1)%max;
        queue[rear]=val;
```

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```
        }
    }
}

void display()
{
    int i;
    if(front==-1)
    {
        printf("empty queue");
    }
    else
    {
        i=front;
        while(1)
        {
            printf("%d ", queue[i]);
            if(i==rear)
            {
                break;
            }
            i=(i+1)%max;
        }
    }
}
```

```
int main()
{
    create_circular_queue();
    display();
    return 0;
}
```

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Enqueue :- To insert a ~~code~~ value from rear.

Algorithm :-

Step 1; IF ((rear + 1) % max == front)

print " queue is full "

Step 2; else if (front == -1)

front ++

Step 3; else,

rear = (rear+1) % max

Step 4; queue[rear] = val;

Step 5; Exit

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Pseudo code for Enqueue :

```
int i,val;
printf("enter value to insert");
scanf("%d",&val);
if((rear+1)%max==front)
{
    printf("queue is full");
}
else if(front==-1)
{
    front++;
}
else
{
    rear=(rear+1)%max;
}
queue[rear]=val;
```

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Dequeue :- To delete a ~~node~~ value from front

Algorithm :-

Step 1:- IF front = = -1
printf " queue is empty"

Step 2:- else IF (front = = rear)
front = rear = -1

Step 3; else,
front = (front +1) % max

Step 4; Exit

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→ Pseudo code for dequeue:

```
if(front == -1)
{
        printf("queue is empty");
}

else if( front == rear)
{
        front = rear = -1;
}
else
{
        front = (front + 1) % max;
}
```

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